

**INDIRA GANDHI NATIONAL OPEN
UNIVERSITY
BACHELOR OF COMPUTER APPLICATIONS
PROJECT REPORT
CROP PREDICTION SYSTEM**

By

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Enrolment No: 198301360

Under the guidance of;

KEERTHANA BALAKRISHNAN



**Submitted to the School of Computer and Information
science in partial fulfillment of the requirements for the
of the degree Bachelor of Computer Applications (BCA)**

**Indira Gandhi National Open University
Maidan Garhi
New Delhi-110068**

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PROFORMA



SCHOOL OF COMPUTER AND INFORMATION SCIENCES
IGNOU, MAIDAN GARHI, NEW DELHI - 110 068

II. PROFORMA OF BCA PROJECT PROPOSAL (BCSP-064) (Project's Title and Guide's Details)

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7. Qualification of the Guide*
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Ph.D.

☐

M.Tech.

☐

B.Tech.

☐

MCA Any

☐

other

☒

*Note : i. All the above mentioned Degrees must have been awarded in Computer Science/IT only ii. A Guide should not guide more than 8 students of BCA at any point of time.

8. Industrial / Teaching experience of the Guide (in Years) 3.4 YEAR

9. Software Used for this Project: PYTHON

Note : 1. Use of Visual Basic and MS-Access as Front End and Back End respectively is forbidden. But, you are permitted to use Visual Basic with other Software. Also, you can use MS-Access with other software.

2. Use of C or C++ Programming Language for Project Related to Database Management is strictly forbidden.

Signature of the Student

Date: 21/12/2021

Signature of the Guide

Date: 20/12/2021

Important:

1. Attach this Proforma along with Guide's Biodata and Project Synopsis in the Project Report.
2. Not more than one student is permitted to work on a project.
3. Complete project as per the comments of Synopsis evaluator, then only submit your Project Report.

For Office Use Only

☒

Approved

☐

Not approved

Signature, Designation, Stamp of the
Project Proposal Evaluator

Date: 13/1/2022

Associate Professor

Department of Computer Applications
Jawahar University of Science & Technology

Suggestions for reformulating the Project:

1. An SRS required.
2. Explain the architecture of the system together with the ML modules you are going to use.
3. Add the details of the dataset you are going to use and its description.

CERTIFICATE OF ORIGINALITY

This is to certify that the project report entitled **CROP PREDICTION SYSTEM** Submitted to Indira Gandhi Open National University in partial fulfilment of the requirement for the award of the degree of BACHELOR OF COMPUTER APPLICATIONS (BCA), is an original work carried out by Mr./Mrs. Varanya Enrolment No. 198301360 under the guidance of Mr./Mrs. Keerthana balakrishnan .

The matter embodied in this project is a genuine work done by the student and has not been submitted whether to this University or to any other University / Institute for the fulfillment of the requirement of any course of study.

Signature of the Student

Signature of the Guide

Varanya c,
Triveny Nilayam,
P. O. Mozhapala, Kannur –
670611 Kerala, India

CERTIFICATE OF COMPLETION

This is to certify that Mr. VARANYA C, Enrolment No: 198301360 have successfully completed Project in BCSP-064 titled “CROP PREDICTION SYSTEM” under the guidance of Mrs. KEERTHANA BALAKRISHNAN .

VARANYA C

Enrolment Number: 198301360

Study Centre: SREE NARAYANA COLLEGE (1405)

Regional Centre: VATAKARA (83)

Place:

Date:

KEERTHANA BALAKRISHNAN

Tutor in Computer Science

Vinodalayam,

Thokkilangadi

Place:

Date:

ACKNOWLEDGEMENT

With proud and pleasure, I submit this project entitled “CROP PREDICTION SYSTEM”. I am very much grateful to the almighty for giving me strength and ability to accomplish this task. I convey my hearty thanks to all the person who helped me in this attempt. I wish to express my heartfelt thanks to my project guide Mrs. **KEERTHANA BALAKRISHNAN** and **ALG international institute of technology** for give me the chance to make this memorable project.

Sincerely

VARANYA C

ABSTRACT

For most developing countries, agriculture is their primary source of revenue. Modern agriculture is a constantly growing approach for agricultural advances and farming techniques. It becomes challenging for the farmers to satisfy our planet's evolving requirements and the expectations of merchants, customers, etc. Some of the challenges the farmers face are-(i) Dealing with climatic changes because of soil erosion and industry emissions (ii) Nutrient deficiency in the soil, caused by a shortage of crucial minerals such as potassium, nitrogen, and phosphorus can result in reduced crop growth. (iii) Farmers make a mistake by cultivating the same crops year after year without experimenting with different varieties. They add fertilizers randomly without understanding the inferior quality or quantity. The paper aims to discover the best model for crop prediction, which can help farmers decide the type of crop to grow based on the climatic conditions and nutrients present in the soil.

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1

INTRODUCTION

1.1. INTRODUCTION

This project is “CROP PREDICTION SYSTEM”. This is an automatic crop prediction system based on weather and soil data

Machine learning is a valuable decision-making tool for predicting agricultural yields and deciding the type of crops to sow and things to do during the crop growing season. In order to aid crop prediction studies, several machine learning methods have been used. Machine learning techniques are utilized in various sectors, from evaluating customer behavior in supermarkets to predicting customer phone usage. For some years, agriculture has been using machine learning techniques. Crop prediction is one of agriculture's complex challenges, and several models have been developed and proven so far. Because crop production is affected by many factors such as atmospheric conditions, type of fertilizer, soil, and seed, this challenge necessitates using several datasets. This implies that predicting agricultural productivity is not a simple process; rather, it entails a series of complicated procedures. Crop yield prediction methods can now reasonably approximate the actual yield, although more excellent yield prediction performance is still desired.

2

OBJECTIVES

2.1. OBJECTIVE OF THE STUDY

The identified challenges motivate to bring up a solution to all the problems stated in the above problem statement section. Following are the objectives of the proposed approach and this thesis work:

➤ **Primary objective**

The proposed system aims at predicting or forecasting the crop yield by learning the past data of the farming land. By considering various factors such as soil conditions, rainfall, temperature, Ph and other entities the system builds a predicting a model using machine learning techniques.

➤ **Secondary Objective**

This, technology will prove to be beneficial to agriculture which will increase crop productivity resulting in better yields to the farmer. Weather forecast data such as temperature and rainfall and soil parameters repository gives insight into which crops are suitable to be cultivated in a particular area.

Purpose

This system provides a platform for right information at right time and also gives a clear idea about agriculture developments such as information on crop, fertilizer, soil. Due to the abnormal weather that happens every year and rapid regional climate change it is difficult to predict the agricultural crop production.

SCOPE OF THE PROJECT

Agriculture is one of the most important occupations practiced in our country. Thus, modernization of agriculture is very important and thus will lead the farmers of our country towards profit. Data analytic (DA) is the process of examining data sets in order to draw conclusions about the information they contain, increasingly with the aid of specialized systems and software. Earlier yield prediction was performed by considering the farmer's experience on a particular field and crop. However, as the conditions change day by day very rapidly, farmers are forced to cultivate more and more crops. Being this as the current situation, many of them don't have enough knowledge about the new crops and are not completely aware of the benefits they get while farming them. Also, the farm productivity can be increased by understanding and forecasting crop performance in a variety of environmental conditions. Thus, the proposed system takes the location of the user as an input. From the location, the nutrients of the soil such as Nitrogen, Phosphorous, Potassium is obtained. The processing part also takes into consideration two more datasets i.e. one obtained from weather department, forecasting the weather expected in current year and the other data being static data. This static data is the crop production and data related to demands of various crops obtained from various government websites. The proposed system applies machine learning and prediction algorithm like Multiple Linear Regression to identify the pattern among data and then process it as per input conditions. This in turn will propose the best feasible crops according to given environmental conditions. Thus, this system will only require the location of the user and it will suggest number of profitable crops providing a choice directly to the farmer about which crop to cultivate. As past year production is also taken into account, the prediction will be more accurate.

3

TOOLS/ENVIRONMENT USED

HARDWARE:-

- **Processor** : **Pentium III**
- **Speed** : **1.1 GHz**
- **Hard Disk** : **20 GB**
- **RAM** : **4 GB**
- **Monitor** : **15 VGA Color**
- **Keyboard** : **Standard Windows Keyboard**

SOFTWARE:-

- **Operating System** : **Windows 7 and above**
- **Back End** : **Python**
- **Tools** : **Python IDE**

4

SYSTEM ANALYSIS

1. Software Requirement Specification

Introduction

In this field all the works are done manually. The main problem of manual entry is, it is time consuming and error prone than the computerized system. There exists wastage of the data retrieval time as in the case of searching and further processing. There was no efficient system to manipulate the existing records.

Purpose

This system provides a platform for right information at right time and also gives a clear idea about agriculture developments such as information on crop, fertilizer, soil. Due to the abnormal weather that happens every year and rapid regional climate change it is difficult to predict the agricultural crop production.

Features

- Automatic crop suggestion based on soil and weather data.
- System predict the crop based on trained data
- By improving the training data , number of predicting crop also we can increase
- System is suitable for new young farmers
- User friendly application

❖ General Description:

This section elaborates the functionality of the intended software in finer details containing subsection. The main function of this software crop prediction from the collection of past data.

Product Perspective:-

This section describes the product interfaces and relationships with other products. It also provides the operational details and business ownership for the products.

Product Function:-

In this section all major functionality of the software product is described. This includes the functional use cases and functionality is normally listed. This software is mainly used for crop prediction

User Characteristics:-

It describes the characteristics of intended software including their background, training requirements, etc.

- User friendly application.
- System suitable for new young farmers

General Constraints

A constraint is something that affects the way that requirements are met. It imposes restrictions on the design of the system that do not affect the external behavior of the system but must be fulfilled to meet technical or project obligations. Typical constraints are time, money, technology, interaction with already existing systems.

Specific Requirements

This section requires the functional and non-functional requirements.

SOFTWARE REQUIREMENTS

HARDWARE:-

- | | | |
|-------------|---|---------------------------|
| • Processor | : | Pentium III |
| • Speed | : | 1.1 GHz |
| • Hard Disk | : | 20 GB |
| • RAM | : | 4 GB |
| • Monitor | : | 15 VGA Color |
| • Keyboard | : | Standard Windows Keyboard |

SOFTWARE:-

- | | | |
|--------------------|---|---------------------|
| • Operating System | : | Windows 7 and above |
| • Back End | : | Python |
| • Tools | : | Python IDE |

2. Software Description

2.1 Android

The official language for Android development is Java. Large parts of Android are written in Java and its APIs are designed to be called primarily from Java. It is possible to develop C and C++ app using the Android Native Development Kit (NDK), however it isn't something that Google promotes. According to Google, "the NDK will not benefit most apps. As a developer, you need to balance its benefits against its drawbacks. Notably, using native code on Android generally does not result in a noticeable performance improvement, but it always increases your app complexity."

Java is a programming language first released by Sun Microsystems back in 1995. It can be found on many different types of devices from smart phones, to mainframe computers. You can use it on your desktop PC and even on the Raspberry Pi. Java doesn't compile to native processor code but rather it relies on a "virtual machine" which understands an intermediate format called Java byte code. Each platform that runs Java needs a virtual machine (VM) implementation. On Android the original VM is called Dalvik. Google has also started previewing its next generation VM called ART. The jobs of these virtual machines are to interpret the byte code, which is really just a set of instructions similar to the machine code found in CPUs, and execute the program on the processor. The VMs use a variety of technologies including just-in-time compilation (JIT) and ahead-of-time compilation (AOT) to speed up the processes.

Android SDK

The Android software development kit (SDK) includes a comprehensive set of development tools. These include a debugger, libraries, a handset emulator based on QEMU, documentation, sample code, and tutorials. Currently supported development platforms include computers running Linux (any modern desktop Linux

distribution), Mac OS X 10.5.8 or later, and Windows 7 or later. As of March 2015, the SDK is not available on Android itself, but software development is possible by using specialized Android applications.

Until around the end of 2014, the officially-supported integrated development environment (IDE) was Eclipse using the Android Development Tools (ADT) Plug-in. As of 2015, Android Studio is the official IDE; however, developers are free to use others, but Google made it clear that ADT was officially deprecated since the end of 2015 to focus on Android Studio as the official Android IDE. Additionally, developers may use any text editor to edit Java and XML files, then use command line tools (Java Development Kit and Apache Ant are required) to create, build and debug Android applications as well as control attached Android devices (e.g., triggering a reboot, installing software package(s) remotely).

2.2 PYTHON

Python is an easy to learn, powerful programming language. It has efficient high-level data structures and a simple but effective approach to object-oriented programming. Python's elegant syntax and dynamic typing, together with its interpreted nature, make it an ideal language for scripting and rapid application development in many areas on most platforms. The Python interpreter and the extensive standard library are freely available in source or binary form for all major platforms from the Python Web site, <https://www.python.org/>, and may be freely distributed. The same site also contains distributions of and pointers to many free third-party Python modules, programs and tools, and additional documentation. The Python interpreter is easily extended with new functions and data types implemented in C or C++ (or other languages callable from C). Python is also suitable as an extension language for customizable applications.

- Python is a popular programming language, and it is one of the most used programming languages today.
- Python works on all the main platforms and operating systems used today, such Windows, Mac OS, and Linux.

- Python is multi-purpose programming languages, which can be used for simulation, creating web pages, communicate with database systems, etc.
- Python is a fairly old Programming Language (1991) compared too many other Programming Languages like C# (2000), Swift (2014), Java (1995), and PHP (1995).
- Python has during the last 10 years become more and more popular. Today,
- Python has become one of the most popular Programming Languages.
- There are many different rankings regarding which programming language which is most popular. In most of these ranking, Python is in top 10.
- One of these rankings is the IEEE Spectrum's ranking of the top programming Languages

Python is an open source, high-level programming language developed by Guido van Rossum in the late 1980s and presently administered by Python Software Foundation. It came from the ABC language that he helped create early on in his career. Python is a powerful language that you can use to create games, write GUIs, and develop web applications. It is a high-level language. Reading and writing codes in Python is much like reading and writing regular English statements. Because they are not written in machine-readable language, Python programs need to be processed before machines can run them. Python is an interpreted language. This means that every time a program is run, its interpreter runs through the code and translates it into machine-readable byte code. Python is an object-oriented language that allows users to manage and control data structures or objects to create and run programs. Everything in Python is, in fact, first class. All objects, data types, functions, methods, and classes take equal position in Python. Programming languages are created to satisfy the needs of programmers and users for an effective tool to develop applications that impact lives, lifestyles, economy, and society. They help make lives better by increasing productivity, enhancing communication, and improving efficiency. Languages die and become obsolete when they fail to live up to expectations and are replaced and superseded by languages that are more powerful. Python is a programming language that has stood the test of time and has remained relevant across industries and businesses and among programmers, and

individual users. It is a living, thriving, and highly useful language that is highly recommended as a first programming language for those who want to dive into and experience programming. Scripting language: A script is a program that controls other programs. Scripting languages are good for quick development and prototyping because they're good at passing messages from one component to another and at handling fiddly stuff like memory management so that the programmer doesn't have to. Python has grown beyond scripting languages, which are used mostly for small applications. The Python community prefers to call Python a dynamic programming language.

Using Python:

- Web applications and frameworks
- Game development
- Data analysis
- Machine learning and artificial intelligence
- GUI based desktop applications
- Enterprise and business applications

Advantages:

- Open source and community development
- Easy to learn
- User-friendly data structures
- High-level language

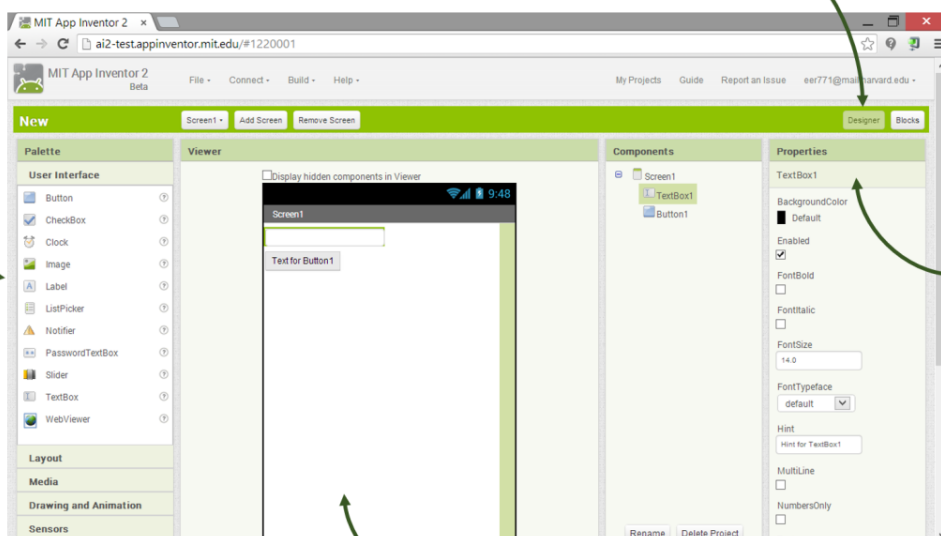
2.3 MIT APP INVENTOR

App Inventor for Android is an open-source web application originally provided by Google, and now maintained by the Massachusetts Institute of Technology (MIT).

It allows newcomers to computer programming to create software applications for the Android operating system (OS). It uses a graphical interface, very similar to Scratch and the Star Logo TNG user interface, which allows users to drag-and-drop visual objects to create an application that can run on Android devices. In creating App Inventor, Google drew upon significant prior research in educational computing, as well as work done within Google on online development environments.

Palette: Find your components and drag them to the Viewer to add them to your app.

Designer Button:
Click from any tab to go to the Designer tab.



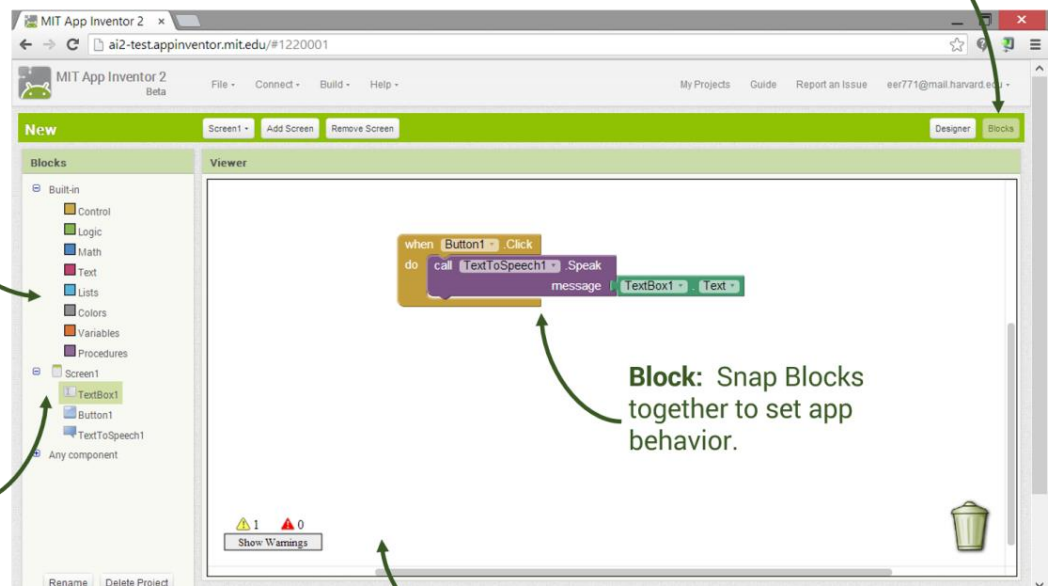
Properties: Select a Component in the Components List to change its properties (color, size, behavior) here.

Viewer: Drag components from the Palette to the Viewer to see what your app will look like.

Built-In Drawers: Find Blocks for general behaviors you may want to add to your app and drag them to the Blocks Viewer.

Blocks Button: Click from any tab to go to the Blocks tab.

Component-Specific Drawers: Find Blocks for behaviors for specific Components and drag them to the Blocks Viewer.



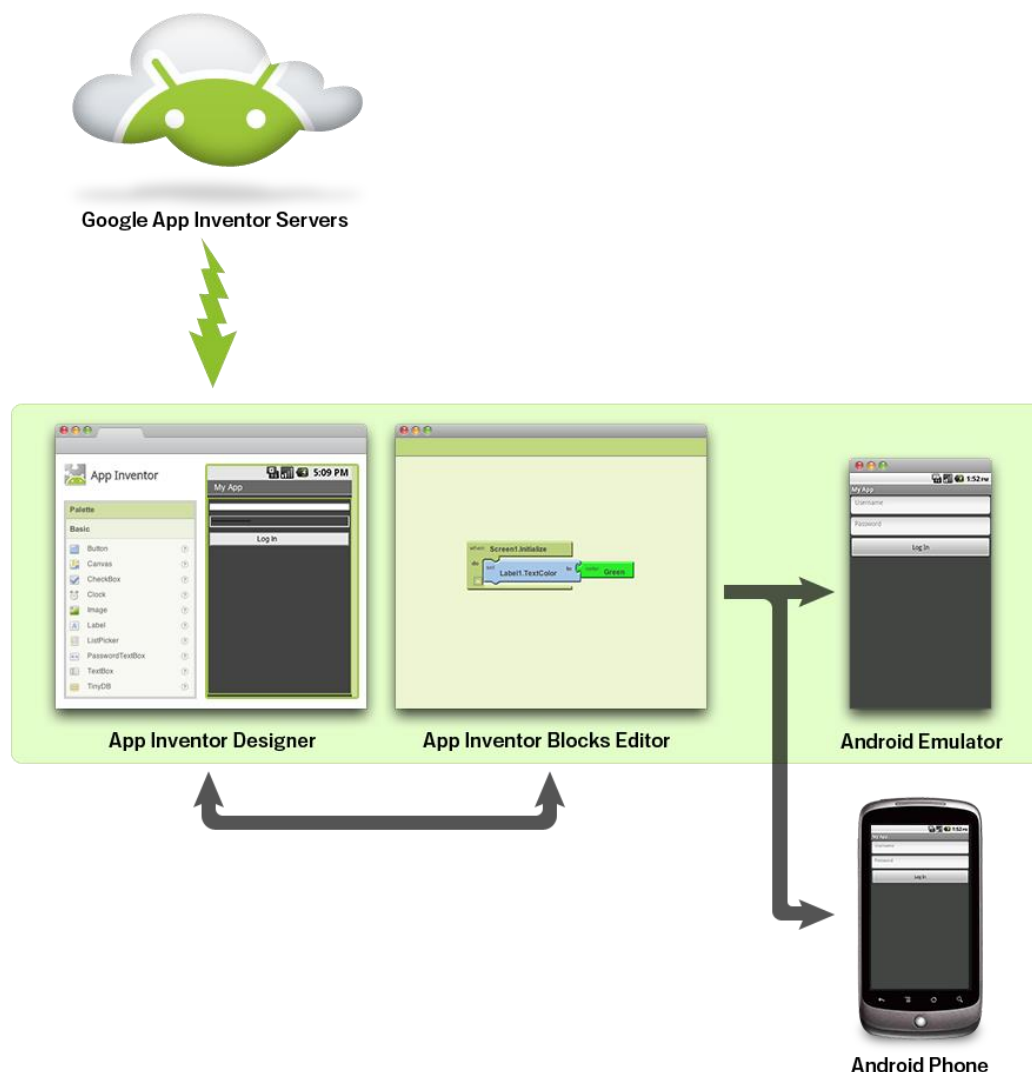
MIT App Inventor is basically for creating Android Apps in your browser where you design how the app will look and function. Like fitting together puzzle pieces, you set how your app will behave to different events by simply signing in with your Gmail Account, so that the App Inventor server can store your work and help you keep track of projects.

It basically consists of two part

1. **App Inventor Designer**
2. **App Inventor Block Editor**

In **App Inventor Designer** you select the components for your app while in **App Inventor Block Editor** you assemble program blocks that specify how the components should behave visually, fitting pieces together like pieces of a puzzle. After completing the above phases you may run your app directly in your Android phone by connecting it to your computer or may run it on Android Emulator if you don't have an Android phone. Moreover you may even download your app (.apk file) and install and run it on your Android device directly.

The picture below shows an overview of how **MIT App Inventor** works!



2.4 FIREBASE

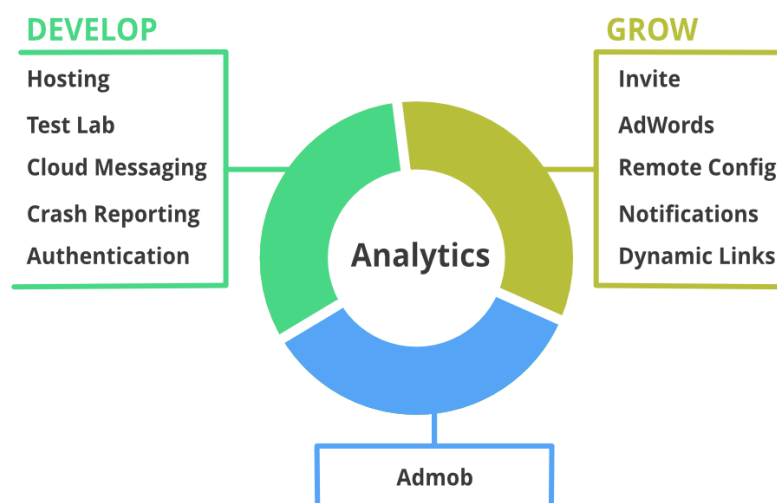
Firebase is a product of Google which helps developers to build, manage, and grow their apps easily. It helps developers to build their apps faster and in a more secure way. No programming is required on the firebase side which makes it easy to use its features more efficiently. It provides services to android, ios, web, and unity. It provides cloud storage. It uses NoSQL for the database for the storage of data.

Brief History of Firebase:

Firebase initially was an online chat service provider to various websites through API and ran with the name Envolve. It got popular as developers used it to exchange application data like a game state in real time across their users more than the chats. This resulted in the separation of the Envolve architecture and its chat system. The Envolve architecture was further evolved by its founders James Templin and Andrew Lee, to what modern day Firebase is in the year 2012.

Features of Firebase:

Mainly there are 3 categories in which firebase provide its services.



Build better applications:

This feature mainly includes backend services that help developers to build and manage their applications in a better way. Services included under this feature are:

- **Real-time Database:** The Firebase Real-time Database is a cloud-based NoSQL database that manages your data at the blazing speed of milliseconds. In simplest term, it can be considered as a big JSON file.
- **Cloud Fire store:** The cloud Fire store is a NoSQL document database that provides services like store, sync, and query through the application on a global scale. It stores data in the form of objects also known as Documents. It has a key-value pair and can store all kinds of data like, strings, binary data, and even JSON trees.
- **Cloud Fire store:** The cloud Fire store is a NoSQL document database that provides services like store, sync, and query through the application on a global scale. It stores data in the form of objects also known as Documents. It has a key-value pair and can store all kinds of data like, strings, binary data, and even JSON trees.
- **Remote Config:** The remote configuration service helps in publishing updates to the user immediately. The changes can range from changing components of the UI to changing the behavior of the applications. These are often used while publishing seasonal offers and contents to the application that has a limited life.
- **Hosting:** Firebase provides hosting of applications with speed and security. It can be used to host Static or Dynamic websites and micro services. It has the capability of hosting an application with a single command.
- **Firebase Cloud Messaging (FCM):** The FCM service provides a connection between the server and the application end users, which can be used to receive and send messages and notifications. These connections are reliable and battery-efficient.

Improve app quality:

Here majorly all the application performance and testing features are provided. All the features required to check and manage before launching your application officially are provided in this section. Services included are:

- **Crashlytics:** It is used to get real-time crash reports. These reports can further be used to improve the quality of the application. The most interesting part of this service is that it gives a detailed description of the crash which is easier to analyze for the developers.
- **Performance monitoring:** This service gives an insight to the performance characteristics of the applications. The performance monitoring SDK can be used to receive performance data from the application, review them, and make changes to the application accordingly through the Firebase console.
- **Test lab:** This service helps to test your applications on real and virtual devices provided by Google which are hosted on the Google Datacenters. It is a cloud-based app-testing infrastructure which supports testing the application on a wide variety of devices and device configurations
- **App Distribution:** This service is used to pre-release applications that can be tested by trusted testers. It comes in handy as decreases the time required to receive feedback from the testers.

Grow your app:

This feature provides your application analytics and features that can help you to interact with your user and make predictions that help you to grow your app. Services provided are:

- **Google analytics:** It is a free app measurement service provided by Google that provides insight on app usage and user engagement. It serves unlimited reporting

for up to 500 distinct automatic or user-defined events using the Firebase SDK.

- **Predictions:** Firebase Predictions uses machine learning to the application's analytics data, further creating dynamic user segments that are based on your user's behavior. These are automatically available to use for the application through Firebase Remote Config, the Notifications composer, Firebase In-App Messaging, and A/B Testing.
- **Dynamic Links:** Deeps Links are links that directly redirects user to specific content. Firebase provides a Dynamic linking service that converts deep links into dynamic links which can directly take the user to a specified content inside the application. Dynamic links are used for converting web users to Native app users. It also increases the conversion of user-to-user sharing. In addition, it can also be used to integrate social media networks, emails, and SMS to increase user engagement inside the application.
- **A/B Testing:** It is used to optimize the application's experience by making it run smoothly, scaling the product, and performing marketing experiments.

Pros and Cons of Using Firebase:

Below listed are the advantages and disadvantages of using a Firebase backend:

- **Pros:**
 - Free plans for beginners.
 - Real-time database is available.
 - Growing Community.
 - Numerous services are available.

- **Cons:**

- It uses NoSQL so, people migrating from SQL might feel difficulty.
- It is still growing so, it is not tested to an extent.

2.5 What is an IDE?

An integrated development environment is a software that helps you write the code. IDE makes developing applications, programs easier. It acts as a text editor, debugger, and compiler. It contains all the tools and features required by a programmer. Hence, it is easier to develop any application or program using an IDE.

2.6 PYTHON IDLE

IDLE stands for Integrated Development and Learning Environment. The story behind the name IDLE is similar to Python. Guido Van Rossum named Python after the British comedy group Monty Python while the name IDLE was chosen to pay tribute to Eric Idle, who was one of the Monty Python's founding members. IDLE comes bundled with the default implementation of the Python language since the 01.5.2b1 release. It is packaged as an optional part of the Python packaging with many Linux, Windows, and Mac distributions.

Python IDLE offers a full-fledged file editor, which gives you the ability to write and execute Python programs from within this program. The built-in file editor also includes several features, like code completion and automatic indentation that will speed up your coding workflow.

IDLE, as shown above, is a very simple and sophisticated IDE developed primarily for beginners, and because of its simplicity, it is highly considered and recommended for educational purposes. It offers a variety of features that you will look in detail along with examples later in this tutorial.

Some of the key features it offers are:

- Python shell with syntax highlighting,
- Multi-window text editor,
- Code auto completion,
- Intelligent indenting,
- Program animation and stepping which allows one line of code to run at a time helpful for debugging,
- Persistent breakpoints

Finally, call stack visibility

2.6 Machine Learning

Machine learning (ML) approaches are used in many fields, ranging from supermarkets to evaluate the behavior of customers to the prediction of customers' phone use. Machine learning is also being used in agriculture for several years . Crop yield prediction is one of the challenging problems in precision agriculture, and many models have been proposed and validated so far. This problem requires the use of several datasets since crop yield depends on many different factors such as climate, weather, soil, use of fertilizer, and seed variety. This indicates that crop yield prediction is not a trivial task; instead, it consists of several complicated steps. Nowadays, crop yield prediction models can estimate the actual yield reasonably, but a better performance in yield prediction is still desirable.

Machine learning, which is a branch of Artificial Intelligence (AI) focusing on learning, is a practical approach that can provide better yield prediction based on several features. Machine learning (ML) can determine patterns and correlations and discover knowledge from datasets. The models need to be trained using datasets, where the outcomes are represented based on past experience. The predictive model is built using several features, and as such, parameters of the models are determined using historical data during the training phase. For the testing phase, part of the historical data that has not been used for training is used for the performance evaluation purpose.

Classification of Machine Learning

Machine learning implementations are classified into four major categories, depending on the nature of the learning “signal” or “response” available to a learning system which is as follows:

1. **Supervised learning:** When an algorithm learns from example data and associated target responses that can consist of numeric values or string labels, such as classes or tags, in order to later predict the correct response when posed with new examples that come under the category of Supervised learning. This approach is indeed similar to human learning under the supervision of a teacher. The teacher provides good examples for the student to memorize, and the student then derives general rules from these specific examples.
2. **Unsupervised learning:** Whereas when an algorithm learns from plain examples without any associated response, leaving the algorithm to determine the data patterns on its own. This type of algorithm tends to restructure the data into something else, such as new features that may represent a class or a new series of un-correlated values. They are quite useful in providing humans with insights into the meaning of data and new useful inputs to supervised machine learning algorithms.

As a kind of learning, it resembles the methods humans use to figure out that

certain objects or events are from the same class, such as by observing the degree of similarity between objects. Some recommendation systems that you find on the web in the form of marketing automation are based on this type of learning.

3. **Reinforcement learning:** When you present the algorithm with examples that lack labels, as in unsupervised learning. However, you can accompany an example with positive or negative feedback according to the solution the algorithm proposes comes under the category of Reinforcement learning, which is connected to applications for which the algorithm must make decisions (so the product is prescriptive, not just descriptive, as in unsupervised learning), and the decisions bear consequences. In the human world, it is just like learning by trial and error. Errors help you learn because they have a penalty added (cost, loss of time, regret, pain, and so on), teaching you that a certain course of action is less likely to succeed than others. An interesting example of reinforcement learning occurs when computers learn to play video games by themselves.

In this case, an application presents the algorithm with examples of specific situations, such as having the gamer stuck in a maze while avoiding an enemy. The application lets the algorithm know the outcome of actions it takes, and learning occurs while trying to avoid what it discovers to be dangerous and to pursue survival. You can have a look at how the company Google DeepMind has created a reinforcement learning program that plays old Atari's video games. When watching the video, notice how the program is initially clumsy and unskilled but steadily improves with training until it becomes a champion.

4. **Semi-supervised learning:** where an incomplete training signal is given: a training set with some (often many) of the target outputs missing. There is a special case of this principle known as Transduction where the entire set of problem instances is known at learning time, except that part of the targets are missing.

About Operating System 2.7

An Operating System may be viewed as an organized collection of software extensions of hardware, consisting of control routines for operating a computer and for providing an environment for execution of programs. Programs rely on facilities provided by the operating system to gain access to computer-system resources such as files and input/output devices. i.e., the operating system acts as interface between users and the hardware of a computer system.

Windows XP is an operating system that uses pre-emptive, multitasking and multi-threading to perform several actions at the same time. The advantage are powerful mobile computing features using the Microsoft Network, a Recycle Bin for storing the deleted files which can be cleared later and the inbox which can be used for communications like FAX, E-MAIL etc. Windows 2000 includes a suite of programs designed to optimize our computer's efficiency, especially when used together. It improves computer reliability by introducing new wizards, utilities, and resources that help keep our system running smoothly. The key features are:

- A very advanced set of multimedia programs are shipped along with the operating system that just doesn't sing or run animations but can create and edit these applications.
- The Operating System automatically detects network cards and user has to just specify the name of the network and the computer can be logged on to the specified network.
- The computer can be connected to the internet if the user has access to it and the operating system needed to log on and run applications on the internet.

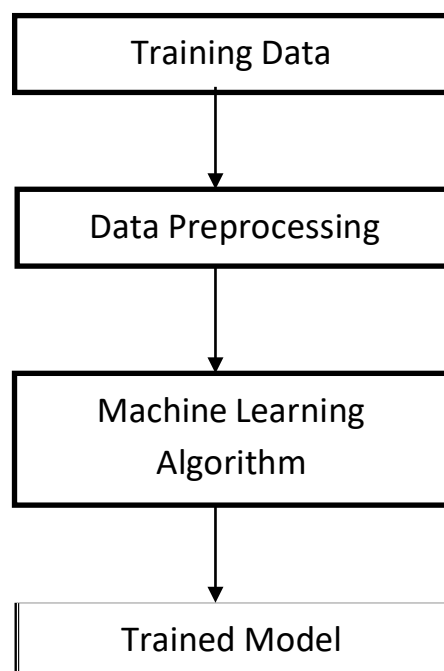
2.8 Analysis Tools

2.8.1 DATA MODELLIING

Data modeling defines primary data objects, composition of each data object, and attributes of the object, relationships between each object and other objects and between objects and the process

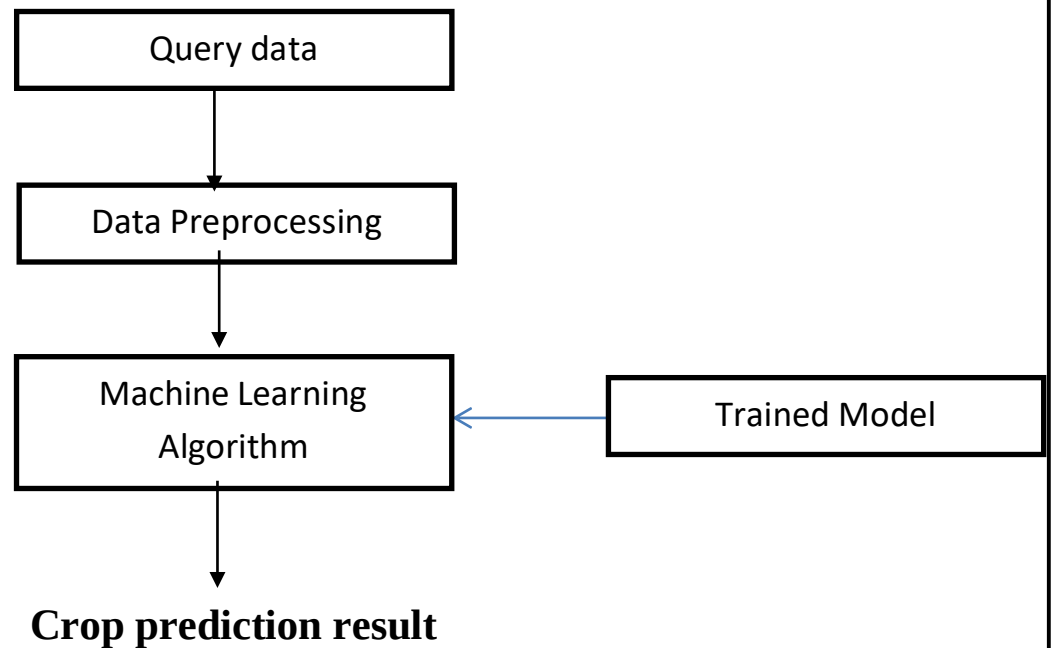
FLOW DIAGRAM :

TRAINING



FLOW DIAGRAM:

TESTING



System architecture represented in the Fig mainly consists of weather API where we fetch the data such as temperature, humidity, rainfall etc. The data fetched from the API are sent to the server module. The data gets stored on to the database on the server. and all these entered data are sent to server. The trained model deployed on the server uses all the fetched and input data for crop yield prediction, finds the yield of predicted crop with its name in the particular area.

5

SYSTEM DESIGN

5.1 MODULES DESIGN

5.2 MODULE DESCRIPTION:-

- TRAINING
- TESTING

TRAINING

- **Training Data :** Soil and weather data

Training data is an extremely large dataset that is used to teach a machine learning model. Training data is used to teach production models that use machine learning algorithms how to extract features that are relevant to specific business goals. For supervised ML models, the training data is labeled. The data used to train unsupervised models is not labeled.

The idea of using training data in machine learning programs is a simple concept, but it is also very foundational to the way that these technologies work. The training data is an initial set of data used to help a program understand how to apply technologies like neural networks to learn and produce sophisticated results. It may be complemented by subsequent sets of data called validation and testing sets.

- **Data Preprocessing :** standardizing the data

Data pre-processing involves transforming raw data to well-formed data sets so that data mining analytics can be applied. Raw data is often incomplete and has inconsistent formatting. The adequacy or inadequacy of data preparation has a direct correlation with the success of any project that involve data analytics.

Pre-processing involves both data validation and data imputation. The goal of data validation is to assess whether the data in question is both complete and accurate. The goal of data imputation is to correct errors and input missing values -- either manually or automatically through Business Processing Automation (BPA) programming.

Data pre-processing is used in both database-driven and rules-based applications. In machine learning (ML) processes, data pre-processing is critical for ensuring large datasets are formatted in such a way that the data they contain can be interpreted and parsed by learning algorithms.

- **Machine Learning Algorithm**

Machine Learning Algorithm: A machine learning algorithm is the method by which the AI system conducts its task, generally predicting output values from given input data. The two main processes of machine learning algorithms are classification and regression.

Machine learning (ML) algorithms are broadly categorized as either supervised or unsupervised. Supervised learning algorithms have both input data and desired output data provided for them through labelling, while unsupervised algorithms work with data that is neither classified nor labelled. An unsupervised algorithm might, for example, group unsorted data according to similarities and differences.

However, many ML approaches, including transfer learning and active learning, involve what are more accurately described as semi-supervised algorithms. Transfer learning uses knowledge gained from completing one task to help solve a different but related problem, while active learning allows an algorithm to query the user or some other source for more information.

- **Trained Model :** save this trained model for testing

A training model is a dataset that is used to train an ML algorithm. It consists of the sample output data and the corresponding sets of input data that have an influence on the output. The training model is used to run the input data through the algorithm to correlate the processed output against the sample output. The result from this correlation is used to modify the model.

This iterative process is called “model fitting”. The accuracy of the training dataset or the validation dataset is critical for the precision of the model.

Model training in machine language is the process of feeding an ML algorithm with data to help identify and learn good values for all attributes involved. There are several types of machine learning models, of which the most common ones are supervised and unsupervised learning.

Supervised learning is possible when the training data contains both the input and output values. Each set of data that has the inputs and the expected output is called a supervisory signal. The training is done based on the deviation of the processed result from the documented result when the inputs are fed into the model.

Unsupervised learning involves determining patterns in the data. Additional data is then used to fit patterns or clusters. This is also an iterative process that improves the accuracy based on the correlation to the expected patterns or clusters. There is no reference output dataset in this method.

TESTING

- **Query data** : testing data

A query is a request for data or information from a database table or combination of tables. This data may be generated as results returned by Structured Query Language (SQL) or as pictorials, graphs or complex results, e.g., trend analyses from data-mining tools.

One of several different query languages may be used to perform a range of simple to complex database queries. SQL, the most well-known and widely-used query language, is familiar to most database administrators (DBAs). A query is a request for data or information from a database table or combination of tables. This data may be generated as results returned by Structured Query Language (SQL) or as pictorials, graphs or complex results, e.g., trend analyses from data-mining tools.

One of several different query languages may be used to perform a range of simple to complex database queries. SQL, the most well-known and widely-used query language, is familiar to most database administrators (DBAs).

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Data pre-processing involves transforming raw data to well-formed data sets so that data mining analytics can be applied. Raw data is often incomplete and has inconsistent formatting. The adequacy or inadequacy of data preparation has a direct correlation with the success of any project that involves data analytics.

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Data pre-processing is used in both database-driven and rules-based applications. In machine learning (ML) processes, data pre-processing is critical for ensuring large datasets are formatted in such a way that the data they contain can be interpreted and parsed by learning algorithms.

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- **Trained Model :** load the trained model

A training model is a dataset that is used to train an ML algorithm. It consists of the sample output data and the corresponding sets of input data that have an influence on the output. The training model is used to run the input data through the algorithm to correlate the processed output against the sample output. The result from this correlation is used to modify the model.

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Supervised learning is possible when the training data contains both the input and output values. Each set of data that has the inputs and the expected output is called a supervisory signal. The training is done based on the deviation of the processed result from the documented result when the inputs are fed into the model.

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- **Crop prediction Result** : recommended crop based on query data

5.3. USER INTERFACE DESIGN

User gets a friendly environment to interact with the software fields. As we are doing the program in PHP and Android the user can view different forms and data access as well as entering is easy. The user’s interaction with system is through forms. Forms are related to databases, so that it becomes easy to enter the data in to the system. Once the input forms are designed the actual data must be entered in to forms, the data entry must be done carefully to avoid any incorrect data. Forms are designed in such a way that during the processing the cursor is placed in the position where the data must be entered the designed forms must be user friendly.

INPUT DESIGN

The goal of input design is to capture data and get that data into a format suitable for the computer. When we computerize a system, the user-oriented program is to be converted into programmer-oriented specification for the computer-based system. All systems need some input data for processing. In order to achieve the objectives of the system, input should be given accurately at the right time. Any ambiguity concerning the inputs will lead to total faults in the functioning of the System.

The most common source of inaccurate data processing errors is the inaccurate input data, the errors made by users can be controlled by input design. The aim of input design is to make operation as easy and error free as possible.

OUTPUT DESIGN

The output design has been done so that the results of processing should be communicated to the user. Effective output design will improve the clarity and performance of outputs. Output is the main reason for developing the system and the basis on which they will evaluate the usefulness of the application.

Output design phase of the system is concerned with the convergence of information to the end user-friendly manner. The output design should be efficient, intelligible so that system relationship with the end user is improved and they're by enhancing the process of decision-making.

NORMALIZATION

The process of normalization is concerned with the transformation of the conceptual scheme to a computer represent able form. Normalization reduces the redundancies and anomalies

The First Normal Form

First normal form does not allow multi valued and composite valued attributes. It states that the domain of an attribute must include only atomic values and that value of any attribute in a tuple must be single value from the domain of that attribute

The second Normal Form

In second normal form, for relation where primary key contains multiple attributes, non-key attributes should be functionally dependent on a part of the primary key.

Third Normal Form

In third normal form, relation should not have a non-key attribute functionally determined by another non-key attribute. That is there should be no transitive dependency of a non-key attribute on the primary key. DE design is required to manage large bodies of information. The management involves both the definition of the structure of storage of information and provisions of mechanism for the manipulation of information

DATASET

5.4 Dataset

Temperature	Humidity	Ph	Rainfall	Seed
20.87974371	82.00274423	6.502985	202.9355362	Rice
21.77046169	80.31964408	7.038096	226.6555374	Rice
31.48468379	50.66105961	6.898284	78.67221693	Wheat
23.22281906	59.31511313	6.277127	108.6591857	Wheat
22.61359953	63.69070564	5.749914	87.75953857	Maize
26.10018422	71.57476937	6.931757	102.266445	Maize
32.82652632	60.63131854	6.192065	87.79470512	Ground nut
33.84404645	61.36211063	6.331147	92.65212945	Groundnut
28.40979139	81.21372954	8.06275	170.3307535	Sugarcane
20.87974371	82.00274423	6.502985	202.9355362	Rice
11.98185482	93.11385534	7.613096	313.0466473	Rice
15.63839393	85.25729183	6.628386	212.3673835	Rice
14.36839494	81.36378479	8.536377	111.6383994	Rice
20.63830473	73.63839393	6.644737	121.6383939	Rice
22.73839393	80.35637304	5.6383399	143.4784993	Rice
43.73839394	86.53637374	3.637383	132.5372828	Rice
14.63738289	79.63829292	2.6373839	123.6373738	Rice
15.64839393	80.63839292	3.3673738	111.6383939	Wheat
32.53738293	76.64829929	4.6483929	121.3673839	Wheat
17.36738292	98.64839393	5.3683839	94.63738383	Wheat
21.53738292	97.53839293	6.3683839	232.6389393	Wheat
24.63839292	67.63839399	7.6373839	45.64839292	Wheat
18.36738299	87.53738283	8.3683929	99.53829202	Wheat
10.73939992	88.63839293	2.3682929	90.53829202	Wheat
19.63738292	98.63839399	5.3627829	121.5383929	Wheat
26.53728281	89.63738299	6.5738393	142.6383929	Maize
29.53728292	90.37939202	4.63738392	132.3682937	Maize
34.24526272	78.73839393	2.36738383	142.6383823	Maize
21.42627272	98.74939303	7.6373838	321.5362729	Maize
31.25628289	90.73829392	8.6373839	243.5362728	Maize
34.53728922	97.73829228	9.74839398	134.5262727	Maize
24.41627829	87.37829292	7.6383939	342.5272728	Maize
32.4268392	91.63738289	3.638383	256.5373839	Maize
52.52738393	93.42267282	5.5363738	70.63939329	Ground nut
24.52637389	89.35738292	7.6373838	374.6373839	Ground nut
21.43738299	79.53278292	4.3783838	453.6373829	Ground nut
24.73897898	32.36829292	8.6484839	567.3572929	Ground nut
26.38678955	76.63828299	2.36733893	134.6373829	Ground nut
24.65498967	89.73839399	9.7383929	234.6382829	Ground nut
23.12357844	98.73838389	1.37838292	123.4567891	Ground nut
24.65846933	98.72829292	6.4738929	435.5272829	Ground nut
40.23412367	123.3782829	9.6382929	213.6373829	Sugarcane
21.53245611	789.738292	4.3783939	452.6382929	Sugarcane
12.67358942	187.278282	5.3663738	213.5372828	Sugarcane
23.34521234	622.7338838	7.6288292	123.6272829	Sugarcane

21.67534467	467.6383939	4.6372882	324.5272829	Sugarcane
26.34789235	321.2527382	6.363839	156.2672829	Sugarcane
21.56734523	132.5262727	8.63728292	163.6382829	Sugarcane
25.35676512	141.2672721	4.52728288	123.6372828	Sugarcane
23.34521234	123.6282829	7.36728289	542.2628829	Sugarcane

Normalized dataset

Temperature	Humidity	PH	Rainfall	Seed
0.243	0.066	0.612	0.301	Rice
0.264	0.063	0.676	0.347	Rice
0.496	0.024	0.659	0.063	Wheat
0.299	0.036	0.585	0.121	Wheat
0.284	0.041	0.522	0.081	Maize
0.368	0.052	0.663	0.109	Maize
0.529	0.037	0.575	0.081	Ground nut
0.553	0.038	0.592	0.09	Groundnut
0.423	0.064	0.799	0.239	Sugarcane
0.243	0.066	0.612	0.301	Rice
0.03	0.08	0.745	0.513	Rice
0.117	0.07	0.627	0.32	Rice
0.087	0.065	0.855	0.126	Rice
0.237	0.054	0.629	0.146	Rice
0.287	0.063	0.509	0.188	Rice
0.79	0.072	0.27	0.167	Rice
0.093	0.062	0.15	0.149	Rice
0.117	0.064	0.238	0.126	Wheat
0.522	0.058	0.391	0.145	Wheat
0.159	0.088	0.477	0.094	Wheat
0.258	0.086	0.596	0.358	Wheat
0.333	0.047	0.748	0	Wheat
0.183	0.073	0.835	0.103	Wheat
0	0.074	0.118	0.086	Wheat
0.213	0.088	0.476	0.145	Wheat
0.378	0.076	0.621	0.186	Maize
0.45	0.077	0.389	0.166	Maize
0.563	0.061	0.118	0.186	Maize
0.256	0.088	0.748	0.529	Maize
0.491	0.077	0.867	0.379	Maize
0.569	0.086	1	0.17	Maize
0.327	0.073	0.748	0.569	Maize
0.519	0.078	0.27	0.404	Maize
1	0.081	0.497	0.048	Ground nut
0.33	0.075	0.748	0.631	Ground nut
0.256	0.062	0.358	0.782	Ground nut
0.335	0	0.869	1	Ground nut
0.374	0.058	0.118	0.171	Ground nut
0.333	0.076	0.999	0.362	Ground nut
0.296	0.088	0	0.149	Ground nut
0.333	0.088	0.609	0.747	Ground nut
0.706	0.12	0.987	0.322	Sugarcane
0.258	1	0.358	0.78	Sugarcane
0.046	0.205	0.476	0.322	Sugarcane
0.302	0.779	0.747 ⁴²	0.149	Sugarcane
0.262	0.575	0.389	0.535	Sugarcane

0.374	0.381	0.596	0.212	Sugarcane
0.259	0.132	0.867	0.226	Sugarcane
0.35	0.144	0.376	0.149	Sugarcane
0.302	0.12	0.716	0.952	Sugarcane

Statistical parameter

PARAMETER	PARAMETER VALUE
Temp Min	10.73939992
Humidity Min	32.36829292
PH Min	1.37838292
Rainfall Min	45.64839292
Temp Max	52.52738393
Humidity Max	789.738292
PH Max	9.74839398
Rainfall Max	567.3572929
Temp Norm Max	41.78798401
Humidity Norm Max	757.3699991
PH Norm Max	8.37001106
Rainfall Norm Max	521.7089

6

PROGRAM CODE

Main program

```
from firebase import Firebase
```

```
import time
```

```
from predict import crop_predict
```

```
config = {
```

```
    "apiKey": "AIzaSyCjM4iy43XcmqJXXm1vLSXiVDKcIJazMCM",
```

```
    "authDomain": "thinkfotechinnovations.firebaseio.com",
```

```
    "databaseURL": "https://thinkfotechinnovations.firebaseio.com",
```

```
    "storageBucket": "thinkfotechinnovations.appspot.com",
```

```
}
```

```
firebase = Firebase(config)
```

```
db = firebase.database()
```

```
storage = firebase.storage()
```

```
#####
```

```
#####
```

```
print("Enter Values for Crop Prediction Through Android Application\n")
```

```
while True:
```

```
    predictionFlag =
```

```
    db.child("CROP_PREDICTION").child("predictionFlag").get().val()
```

```
    if predictionFlag == None:
```

```
        predictionFlag = "0"
```

```
        db.child("CROP_PREDICTION").child("predictionFlag").set(predictionFlag)
```

```
#####
```

```
#####
```

```
    if predictionFlag == "1":
```

```
        Temperature =
```

```
        db.child("CROP_PREDICTION").child("temperature").get().val()
```

```
        Humidity = db.child("CROP_PREDICTION").child("humidity").get().val()
```

```
        Ph = db.child("CROP_PREDICTION").child("pH").get().val()
```

```
        Rainfall = db.child("CROP_PREDICTION").child("rainfall").get().val()
```

```
        predictionFlag = "0"
```

```
db.child("CROP_PREDICTION").child("predictionFlag").set(predictionFlag)
```

```
#####
```

```
#####
```

```
PredictionStatus = crop_predict(Temperature, Humidity, Ph, Rainfall)
```

```
print("\nPredicted Result: ", PredictionStatus, "\n\n")
```

```
PredictionStatus = PredictionStatus.replace(" ", "*")
```

```
db.child("CROP_PREDICTION").child("result").set(PredictionStatus)
```

```
db.child("CROP_PREDICTION").child("resultFlag").set("1")
```

Prediction program

```
# importing openpyxl module
```

```
import openpyxl
```

```
from firebase import Firebase
```

```
config = {
```

```
    "apiKey": "AIzaSyCjM4iy43XcmqJXXm1vLSXiVDKcIJazMCM",
```

```
    "authDomain": "thinkfotechinnovations.firebaseio.com",
```

```
    "databaseURL": "https://thinkfotechinnovations.firebaseio.com",
```

```
"storageBucket": "thinkfotechinnovations.appspot.com",  
}
```

```
firebase = Firebase(config)
```

```
db = firebase.database()
```

```
storage = firebase.storage()
```

```
def crop_predict(temp_query, humidity_query, ph_query, rainfall_query):
```

```
    temp_query = float(temp_query)
```

```
    humidity_query = float(humidity_query)
```

```
    ph_query = float(ph_query)
```

```
    rainfall_query = float(rainfall_query)
```

```
    path2 = "Statistical_Parameters.xlsx"
```

```
    # workbook object is created
```

```
    wb_obj2 = openpyxl.load_workbook(path2)
```



```
sheet_obj2 = wb_obj2.active
```

```
m_row2 = sheet_obj2.max_row
```

```
#####
```

```
#####
```

```
c5 = sheet_obj2.cell(row=2, column=2)
```

```
temp_min = c5.value
```

```
c6 = sheet_obj2.cell(row=3, column=2)
```

```
humidity_min = c6.value
```

```
c7 = sheet_obj2.cell(row=4, column=2)
```

```
ph_min = c7.value
```

```
c8 = sheet_obj2.cell(row=5, column=2)
```

```
rainfall_min = c8.value
```

```
#print(temp_min, humidity_min, ph_min, rainfall_min)
```

#####

#####

```
c9 = sheet_obj2.cell(row=10, column=2)
```

```
norm_temp_max = c9.value
```

```
c10 = sheet_obj2.cell(row=11, column=2)
```

```
norm_humidity_max = c10.value
```

```
c11 = sheet_obj2.cell(row=12, column=2)
```

```
norm_ph_max = c11.value
```

```
c12 = sheet_obj2.cell(row=13, column=2)
```

```
norm_rainfall_max = c12.value
```

```
#print("\n", norm_temp_max, norm_humidity_max, norm_ph_max,  
norm_rainfall_max)
```

#####

```
#####
```

```
c13 = sheet_obj2.cell(row=6, column=2)
```

```
temp_max = c13.value
```

```
c14 = sheet_obj2.cell(row=7, column=2)
```

```
humidity_max = c14.value
```

```
c15 = sheet_obj2.cell(row=8, column=2)
```

```
ph_max = c15.value
```

```
c16 = sheet_obj2.cell(row=9, column=2)
```

```
rainfall_max = c16.value
```

```
#print("\n", temp_max, humidity_max, ph_max, rainfall_max)
```

```
#####
```

```
#####
```

```
if(temp_query < temp_min):  
    Temp_SD = 0  
elif(temp_query > temp_max):  
    Temp_SD = 1  
elif(temp_min <= temp_query <= temp_max):  
    Temp_SD = (temp_query - temp_min)/norm_temp_max  
print("Temp_SD: ", Temp_SD)
```

```
if(humidity_query < humidity_min):  
    Humidity_SD = 0  
elif(humidity_query > humidity_max):  
    Humidity_SD = 1  
elif(humidity_min <= humidity_query <= humidity_max):  
    Humidity_SD = (humidity_query - humidity_min)/norm_humidity_max  
print("Humidity_SD: ", Humidity_SD)
```

```
if(ph_query < ph_min):  
    PH_SD = 0  
elif(ph_query > ph_max):  
    PH_SD = 1
```

```
elif(ph_min <= ph_query <= ph_max):
```

```
    PH_SD = (ph_query - ph_min)/norm_ph_max
```

```
print("PH_SD: ", PH_SD)
```

```
if(rainfall_query < rainfall_min):
```

```
    Rainfall_SD = 0
```

```
elif(rainfall_query > rainfall_max):
```

```
    Rainfall_SD = 1
```

```
elif(rainfall_min <= rainfall_query <= rainfall_max):
```

```
    Rainfall_SD = (rainfall_query - rainfall_min)/norm_rainfall_max
```

```
print("Rainfall_SD: ", Rainfall_SD)
```

```
#####
```

```
#####
```

```
#####
```

```
Temp_SD_list = []
```

```
Humidity_SD_list = []
```

```
PH_SD_list = []
```

```
Rainfall_SD_list = []
```

```
seed_list = []
```

```
EDr_list = []
```

```
path3 = "Normalized_Dataset.xlsx"
```

```
# workbook object is created
```

```
wb_obj3 = openpyxl.load_workbook(path3)
```

```
sheet_obj3 = wb_obj3.active
```

```
m_row3 = sheet_obj3.max_row
```

```
for i in range(2, m_row3 + 1):
```

```
    cell_obj0 = sheet_obj3.cell(row = i, column = 5)
```

```
    #print(cell_obj.value)
```

```
    seed_list.append(cell_obj0.value)
```

```
for i in range(2, m_row3 + 1):
```

```
    cell_obj1 = sheet_obj3.cell(row = i, column = 1)
```

```
    Temp_SD_Value = round(pow((Temp_SD - cell_obj1.value),2),3)
```

```
    #print(cell_obj.value)
```

```
    Temp_SD_list.append(Temp_SD_Value)
```

```
#print(Temp_SD_list)
```

```
for i in range(2, m_row3 + 1):
```

```
    cell_obj2 = sheet_obj3.cell(row = i, column = 2)
```

```
    Humidity_SD_Value = round(pow((Humidity_SD - cell_obj2.value),2),3)
```

```
    #print(cell_obj2.value)
```

```
    Humidity_SD_list.append(Humidity_SD_Value)
```

```
#print(Humidity_SD_list)
```

```
for i in range(2, m_row3 + 1):
```

```
    cell_obj3 = sheet_obj3.cell(row = i, column = 3)
```

```
    PH_SD_Value = round(pow((PH_SD - cell_obj3.value),2),3)
```

```
    #print(cell_obj2.value)
```

```
    PH_SD_list.append(PH_SD_Value)
```

```
#print(PH_SD_list)
```

```
for i in range(2, m_row3 + 1):
```

```
    cell_obj4 = sheet_obj3.cell(row = i, column = 4)
```

```
    Rainfall_SD_Value = round(pow((Rainfall_SD - cell_obj4.value),2),3)
```

```

#print(cell_obj2.value)

Rainfall_SD_list.append(Rainfall_SD_Value)


#print(Rainfall_SD_list)


#print(len(Rainfall_SD_list), len(PH_SD_list), len(Humidity_SD_list),
len(Temp_SD_list))


for j in range(0,len(Temp_SD_list)):


    EDr_Value = round((Temp_SD_list[j] + Humidity_SD_list[j] + PH_SD_list[j] +
Rainfall_SD_list[j]),3)

    EDr_list.append(EDr_Value)


#print("\n\n", EDr_list)


EDr_min = min(EDr_list)


print("EDr Min", EDr_min)

```

```
#####
```


#####

#####

```
wb = openpyxl.Workbook()
```

```
wb_new = wb.active
```

```
wb_new.title = "Feature Error Table"
```

```
wb.save(filename = 'Feature_Error.xlsx')
```

```
path4 = "Feature_Error.xlsx"
```

```
# workbook object is created
```

```
wb_obj4 = openpyxl.load_workbook(path4)
```

```
sheet_obj4 = wb_obj4.active
```

```
m_row4 = sheet_obj4.max_row
```

```
CT_1 = sheet_obj4.cell(row=1, column=1)
```

```
CT_1.value = 'Target'
```

```
CT_2 = sheet_obj4.cell(row=1, column=2)
```

```
CT_2.value = 'Seed'
```

```
row_value = 2

for j in EDr_list:

    c1 = sheet_obj4.cell(row=row_value, column=1)

    c1.value = j

    row_value = row_value+1
```

```
row_value = 2

for j in seed_list:

    c2 = sheet_obj4.cell(row=row_value, column=2)

    c2.value = j

    row_value = row_value+1
```

```
wb_obj4.save('Feature_Error.xlsx')
```

```
#####

#####

#####
```

```
l = 0

for row in sheet_obj4.rows:
```

```
if(row[0].value == EDr_min):  
  
    #print(row[1].value)  
  
    db.child("CROP_PREDICTION").child("result").set(row[1].value)  
  
    return(row[1].value)
```

```
l = l + 1
```

Dataset program

```
# importing openpyxl module
```

```
import openpyxl
```

```
# Give the location of the file
```

```
path1 = "dataset.xlsx"
```

```
# workbook object is created
```

```
wb_obj1 = openpyxl.load_workbook(path1)
```

```
sheet_obj1 = wb_obj1.active
```

```
m_row1 = sheet_obj1.max_row
```

```
""wb = openpyxl.Workbook()

wb_new = wb.active

wb_new.title = "Normalized Feature Table"

wb.save(filename = 'Normalized.xlsx')""
```

```
temp_list = []

humidity_list = []

ph_list = []

rainfall_list = []

seed_list = []
```

```
temp_norm_trail_list = []

humidity_norm_trail_list = []

ph_norm_trail_list = []

rainfall_norm_trail_list = []
```

```
temp_norm_list = []

humidity_norm_list = []

ph_norm_list = []

rainfall_norm_list = []
```

```
print("No of rows: ", m_row1, "\n")
```

```
#####
```

```
#####
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 1)
```

```
    #print(cell_obj1.value)
```

```
    temp_list.append(cell_obj1.value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 2)
```

```
    #print(cell_obj1.value)
```

```
    humidity_list.append(cell_obj1.value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 3)
```

```
    #print(cell_obj1.value)
```

```
    ph_list.append(cell_obj1.value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 4)
```

```
    #print(cell_obj.value)
```

```
    rainfall_list.append(cell_obj1.value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 5)
```

```
    #print(cell_obj.value)
```

```
    seed_list.append(cell_obj1.value)
```

```
'''print(temp_list)
```

```
print(humidity_list)
```

```
print(ph_list)
```

```
print(rainfall_list)'''
```

```
temp_min = min(temp_list)
```

```
humidity_min = min(humidity_list)
```

```
ph_min = min(ph_list)
```

```
rainfall_min = min(rainfall_list)
```

```
temp_max = max(temp_list)
```

```
humidity_max = max(humidity_list)
```

```
ph_max = max(ph_list)
```

```
rainfall_max = max(rainfall_list)
```

```
print("Temperature Min: ", temp_min)
```

```
print("Temperature Max: ", temp_max)
```

```
print("\nHumidity Min: ", humidity_min)
```

```
print("Humidity Max: ", humidity_max)
```

```
print("\nPH Min: ", ph_min)
```

```
print("PH Max: ", ph_max)
```

```
print("\nRainfall Min: ", rainfall_min)
```

```
print("Rainfall Max: ", rainfall_max)
```

```
print("\n")
```

#####

#####

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 1)
```

```
    temp_norm_trail_value = (cell_obj1.value - temp_min)
```

```
    temp_norm_trail_list.append(temp_norm_trail_value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 2)
```

```
    humidity_norm_trail_value = (cell_obj1.value - humidity_min)
```

```
    humidity_norm_trail_list.append(humidity_norm_trail_value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 3)
```

```
    ph_norm_trail_value = (cell_obj1.value - ph_min)
```

```
    ph_norm_trail_list.append(ph_norm_trail_value)
```

```
for i in range(2, m_row1 + 1):
```

```
    cell_obj1 = sheet_obj1.cell(row = i, column = 4)
```



```
rainfall_norm_trail_value = (cell_obj1.value - rainfall_min)

rainfall_norm_trail_list.append(rainfall_norm_trail_value)
```

```
norm_temp_max = max(temp_norm_trail_list)

norm_humidity_max = max(humidity_norm_trail_list)

norm_ph_max = max(ph_norm_trail_list)

norm_rainfall_max = max(rainfall_norm_trail_list)
```

```
#####

#####
```

```
for i in temp_norm_trail_list:

    temp_norm_value = round(i/norm_temp_max,3)

    temp_norm_list.append(temp_norm_value)
```

```
for i in humidity_norm_trail_list:

    humidity_norm_value = round(i/norm_humidity_max,3)

    humidity_norm_list.append(humidity_norm_value)
```

```
for i in ph_norm_trail_list:
```

```
    ph_norm_value = round(i/norm_ph_max,3)
```

```
    ph_norm_list.append(ph_norm_value)
```

```
for i in rainfall_norm_trail_list:
```

```
    rainfall_norm_value = round(i/norm_rainfall_max,3)
```

```
    rainfall_norm_list.append(rainfall_norm_value)
```

```
"""print(temp_norm_list)
```

```
print(humidity_norm_list)
```

```
print(ph_norm_list)
```

```
print(rainfall_norm_list)"""
```

```
#####
```

```
#####
```

```
wb = openpyxl.Workbook()
```

```
wb_new = wb.active
```

```
wb_new.title = "Normalized Feature Table"
```

```
wb.save(filename = 'Normalized_Dataset.xlsx')
```

```
path2 = "Normalized_Dataset.xlsx"
```

```
# workbook object is created
```

```
wb_obj2 = openpyxl.load_workbook(path2)
```

```
sheet_obj2 = wb_obj2.active
```

```
m_row2 = sheet_obj2.max_row
```

```
c1 = sheet_obj2.cell(row=1, column=1)
```

```
c1.value = 'Temperature'
```

```
c2 = sheet_obj2.cell(row=1, column=2)
```

```
c2.value = 'Humidity'
```

```
c3 = sheet_obj2.cell(row=1, column=3)
```

```
c3.value = 'PH'
```

```
c4 = sheet_obj2.cell(row=1, column=4)
```

```
c4.value = 'Rainfall'
```

```
c5 = sheet_obj2.cell(row=1, column=5)
```

```
c5.value = 'Seed'
```

```
row_value = 2
```

```
for j in temp_norm_list:
```

```
    c1 = sheet_obj2.cell(row=row_value, column=1)
```

```
    c1.value = j
```

```
    row_value = row_value+1
```

```
row_value = 2
```

```
for j in humidity_norm_list:
```

```
    c1 = sheet_obj2.cell(row=row_value, column=2)
```

```
    c1.value = j
```

```
    row_value = row_value+1
```

```
row_value = 2
```

```
for j in ph_norm_list:
```

```
    c1 = sheet_obj2.cell(row=row_value, column=3)
```

```
    c1.value = j
```

```
    row_value = row_value+1
```

```
row_value = 2
```

```
for j in rainfall_norm_list:
```

```
    c1 = sheet_obj2.cell(row=row_value, column=4)
```

```

c1.value = j

row_value = row_value+1


row_value = 2

for j in seed_list:

    c1 = sheet_obj2.cell(row=row_value, column=5)

    c1.value = j

    row_value = row_value+1

```

```

wb_obj2.save('Normalized_Dataset.xlsx')

```

```

#####

#####

```

```

wb = openpyxl.Workbook()

wb_new = wb.active

wb_new.title = "Statistical Parameter Table"

wb.save(filename = 'Statistical_Parameters.xlsx')


path3 = "Statistical_Parameters.xlsx"

```

```

# workbook object is created

wb_obj3 = openpyxl.load_workbook(path3)

sheet_obj3 = wb_obj3.active

m_row3 = sheet_obj3.max_row

c1 = sheet_obj3.cell(row=1, column=1)

c1.value = 'PARAMETER'

c2 = sheet_obj3.cell(row=1, column=2)

c2.value = 'PARAMETER VALUE'

#####

c1_1 = sheet_obj3.cell(row=2, column=1)

c1_1.value = "Temp Min"

c2_1 = sheet_obj3.cell(row=2, column=2)

c2_1.value = temp_min

c1_2 = sheet_obj3.cell(row=3, column=1)

c1_2.value = "Humidity Min"

c2_2 = sheet_obj3.cell(row=3, column=2)

```

```
c2_2.value = humidity_min
```

```
c1_3 = sheet_obj3.cell(row=4, column=1)
```

```
c1_3.value = "PH Min"
```

```
c2_3 = sheet_obj3.cell(row=4, column=2)
```

```
c2_3.value = ph_min
```

```
c1_4 = sheet_obj3.cell(row=5, column=1)
```

```
c1_4.value = "Rainfall Min"
```

```
c2_4 = sheet_obj3.cell(row=5, column=2)
```

```
c2_4.value = rainfall_min
```

```
#####
```

```
c1_5 = sheet_obj3.cell(row=6, column=1)
```

```
c1_5.value = "Temp Max"
```

```
c2_5 = sheet_obj3.cell(row=6, column=2)
```

```
c2_5.value = temp_max
```

```
c1_6 = sheet_obj3.cell(row=7, column=1)
```

```
c1_6.value = "Humidity Max"
```

```
c2_6 = sheet_obj3.cell(row=7, column=2)
```

```
c2_6.value = humidity_max
```

```
c1_7 = sheet_obj3.cell(row=8, column=1)
```

```
c1_7.value = "PH Max"
```

```
c2_7 = sheet_obj3.cell(row=8, column=2)
```

```
c2_7.value = ph_max
```

```
c1_8 = sheet_obj3.cell(row=9, column=1)
```

```
c1_8.value = "Rainfall Max"
```

```
c2_8 = sheet_obj3.cell(row=9, column=2)
```

```
c2_8.value = rainfall_max
```

```
#####
```

```
c1_9 = sheet_obj3.cell(row=10, column=1)
```

```
c1_9.value = "Temp Norm Max"
```

```
c2_9 = sheet_obj3.cell(row=10, column=2)
```

```
c2_9.value = norm_temp_max
```

```
c1_10 = sheet_obj3.cell(row=11, column=1)
```

```
c1_10.value = "Humidity Norm Max"
```

```
c2_10 = sheet_obj3.cell(row=11, column=2)
```



```
c2_10.value = norm_humidity_max
```

```
c1_11 = sheet_obj3.cell(row=12, column=1)
```

```
c1_11.value = "PH Norm Max"
```

```
c2_11 = sheet_obj3.cell(row=12, column=2)
```

```
c2_11.value = norm_ph_max
```

```
c1_12 = sheet_obj3.cell(row=13, column=1)
```

```
c1_12.value = "Rainfall Norm Max"
```

```
c2_12 = sheet_obj3.cell(row=13, column=2)
```

```
c2_12.value = norm_rainfall_max
```

```
#####
```

```
wb_obj3.save('Statistical_Parameters.xlsx')
```

7

TESTING

7.1. SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub-assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the

Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests

Demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input	:	identified classes of valid input must be accepted.
Invalid Input	:	identified classes of invalid input must be rejected.
Functions	:	identified functions must be exercised
Output	:	identified classes of application outputs must be exercised.
Systems/Procedures	:	interfacing systems or procedures Must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete,

Additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration-oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

7.1 Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

7.2 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

7.3 Acceptance Testing

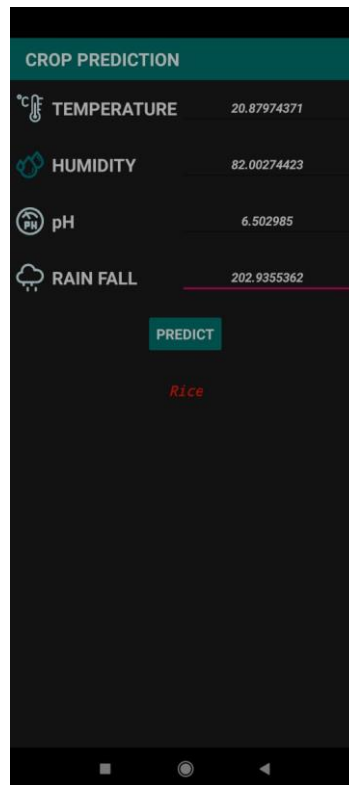
User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

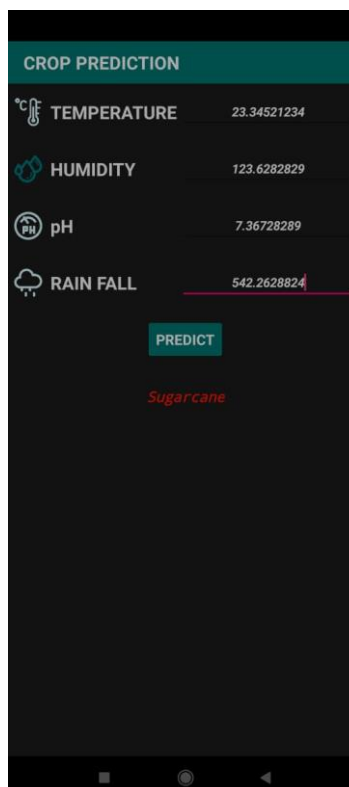
8

SCREENSHOTS

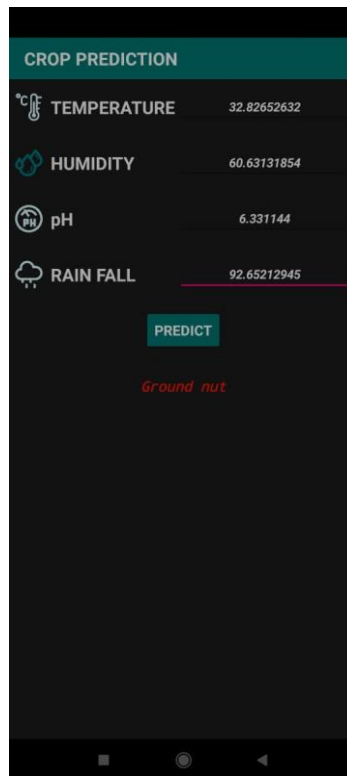
Rice result



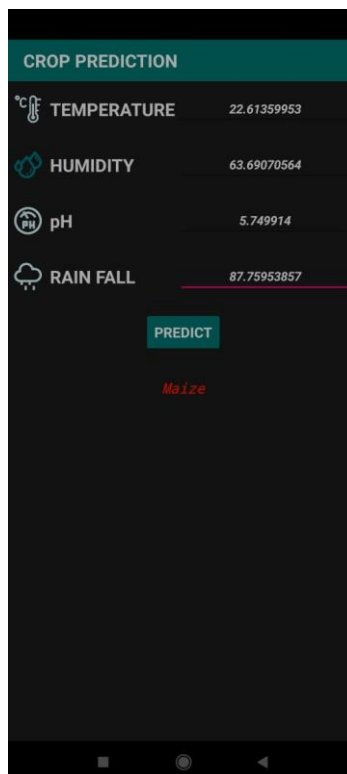
Sugarcane result



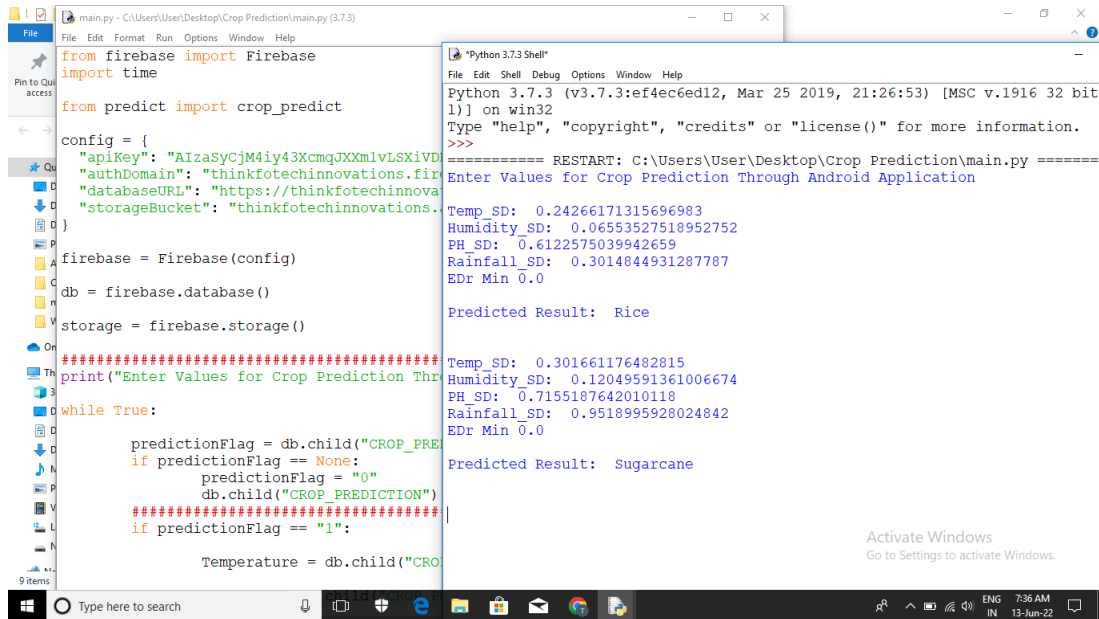
Ground nut result



Maze result



Sugarcane prediction



The screenshot shows a Python script in a text editor and its execution in a Python 3.7.3 Shell. The script uses Firebase to store and retrieve crop prediction data. The output shows two successful predictions: Rice and Sugarcane, each with associated sensor data (Temp_SD, Humidity_SD, PH_SD, Rainfall_SD, EDr Min).

```
main.py - C:\Users\User\Desktop\Crop Prediction\main.py (3.7.3)
File Edit Format Run Options Window Help
from firebase import Firebase
import time

from predict import crop_predict

config = {
    "apiKey": "AIzaSyCjM4iy43XcmqJXXm1vLSx1VD",
    "authDomain": "thinkfotechinnovations.firebaseio.com",
    "databaseURL": "https://thinkfotechinnovations.firebaseio.com",
    "storageBucket": "thinkfotechinnovations.firebaseio.com"
}

firebase = Firebase(config)
db = firebase.database()
storage = firebase.storage()

#####
print("Enter Values for Crop Prediction Through Android Application")

While True:

    predictionFlag = db.child("CROP_PREDICTION").child("predictionFlag").get()
    if predictionFlag == None:
        predictionFlag = "0"
        db.child("CROP_PREDICTION").child("predictionFlag").set(predictionFlag)
    #####
    if predictionFlag == "1":
        Temperature = db.child("CROP_PREDICTION").child("Temperature").get()
        Temp_SD = 0.24266171315696983
        Humidity_SD = 0.06553527518952752
        PH_SD = 0.6122575039942659
        Rainfall_SD = 0.3014844931287787
        EDr Min 0.0

        Predicted Result: Rice

    if predictionFlag == "2":
        Temperature = db.child("CROP_PREDICTION").child("Temperature").get()
        Temp_SD = 0.301661176482815
        Humidity_SD = 0.12049591361006674
        PH_SD = 0.7155187642010118
        Rainfall_SD = 0.9518995928024842
        EDr Min 0.0

        Predicted Result: Sugarcane
```

Python 3.7.3 Shell

```
Python 3.7.3 (v3.7.3:ef4ec6ed12, Mar 25 2019, 21:26:53) [MSC v.1916 32 bit
1)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\User\Desktop\Crop Prediction\main.py =====
Enter Values for Crop Prediction Through Android Application
Temp_SD: 0.24266171315696983
Humidity_SD: 0.06553527518952752
PH_SD: 0.6122575039942659
Rainfall_SD: 0.3014844931287787
EDr Min 0.0

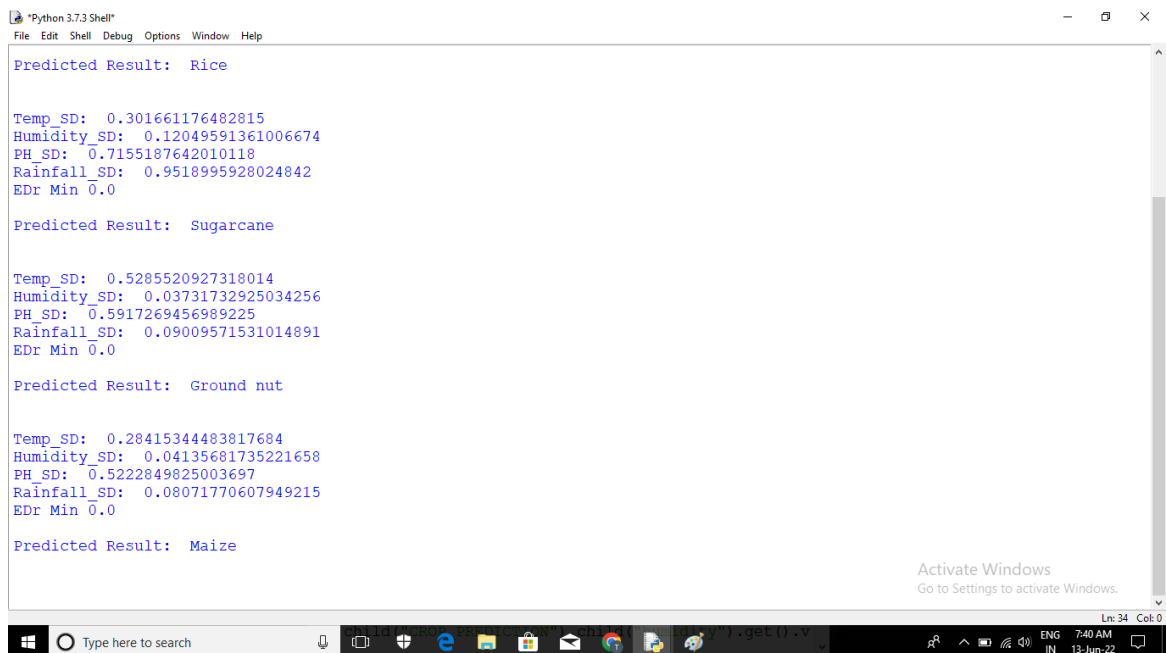
Predicted Result: Rice

Temp_SD: 0.301661176482815
Humidity_SD: 0.12049591361006674
PH_SD: 0.7155187642010118
Rainfall_SD: 0.9518995928024842
EDr Min 0.0

Predicted Result: Sugarcane
```

Activate Windows
Go to Settings to activate Windows.

Maze prediction



The screenshot shows the execution output of a Python script in a Python 3.7.3 Shell. The output displays four successful predictions: Rice, Sugarcane, Ground nut, and Maize, each with associated sensor data (Temp_SD, Humidity_SD, PH_SD, Rainfall_SD, EDr Min).

```
Python 3.7.3 Shell
File Edit Shell Debug Options Window Help

Predicted Result: Rice

Temp_SD: 0.301661176482815
Humidity_SD: 0.12049591361006674
PH_SD: 0.7155187642010118
Rainfall_SD: 0.9518995928024842
EDr Min 0.0

Predicted Result: Sugarcane

Temp_SD: 0.5285520927318014
Humidity_SD: 0.03731732925034256
PH_SD: 0.5917269456989225
Rainfall_SD: 0.09009571531014891
EDr Min 0.0

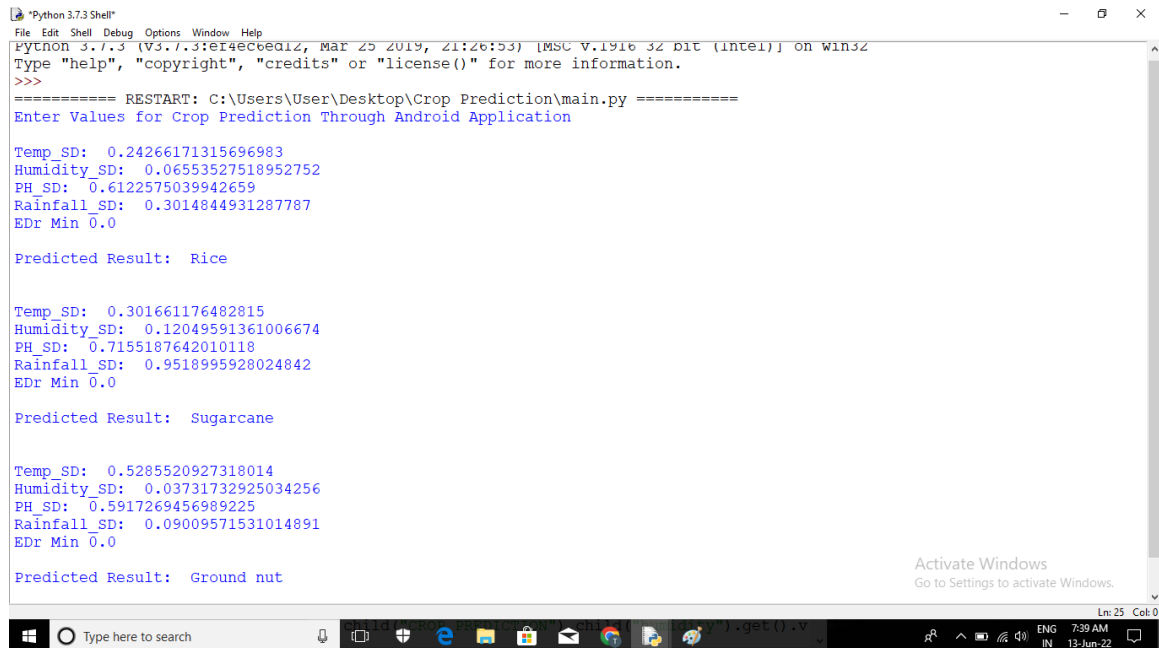
Predicted Result: Ground nut

Temp_SD: 0.28415344483817684
Humidity_SD: 0.04135681735221658
PH_SD: 0.5222849825003697
Rainfall_SD: 0.08071770607949215
EDr Min 0.0

Predicted Result: Maize
```

Activate Windows
Go to Settings to activate Windows.

Groundnut prediction



```
Python 3.7.3 Shell
File Edit Shell Debug Options Window Help
Python 3.7.3 (tags/v3.7.3:ef4ec6ed12, Mar 25 2019, 21:26:53) [MSC v.1916 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
===== RESTART: C:\Users\User\Desktop\Crop Prediction\main.py =====
Enter Values for Crop Prediction Through Android Application

Temp_SD: 0.24266171315696983
Humidity_SD: 0.06553527518952752
PH_SD: 0.6122575039942659
Rainfall_SD: 0.3014844931287787
EDr Min 0.0

Predicted Result: Rice

Temp_SD: 0.301661176482815
Humidity_SD: 0.12049591361006674
PH_SD: 0.7155187642010118
Rainfall_SD: 0.9518995928024842
EDr Min 0.0

Predicted Result: Sugarcane

Temp_SD: 0.5285520927318014
Humidity_SD: 0.03731732925034256
PH_SD: 0.5917269456989225
Rainfall_SD: 0.09009571531014891
EDr Min 0.0

Predicted Result: Ground nut
```

Activate Windows
Go to Settings to activate Windows.

Ln: 25 Col: 0

9. FUTURE APPLICATION

In coming years, can try applying data independent system. That is whatever be the format our system should work with same accuracy. Integrating soil details to the system is an advantage, as for the selection of crops knowledge on soil is also a parameter. Proper irrigation is also needed feature crop cultivation. In reference to rainfall can depict whether extra water availability is needed or not. This research work can be enhanced to higher level by availing it to whole India.

10. LIMITATIONS OF THE PROJECT

- Data needed for crop model calibration and simulation include climatic data, such as precipitation, maximum and minimum, temperature and relative humidity.

11. REFERNCE

- WIKIPEDIA - https://en.wikipedia.org/wiki/Main_Page
- GOOGLE - <https://www.google.com/>
- GEEKSFORGEEKS- <https://www.geeksforgeeks.org/>